

Stochastic Contextual Bandits / Part 2

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Linear bandits

(a generalization of finite bandits)

Setting: At each round $t = 1, 2, \dots$,

0. A decision set $\mathcal{D}_t \subset \mathbb{R}^d$ is revealed
1. Statistician picks $D_t \in \mathcal{D}_t$
2. She gets a reward $Y_t = \langle \theta^*, D_t \rangle + \eta_t$
where η_t is an independent noise
3. This reward Y_t is the **only feedback** she receives

→ **Exploration–exploitation dilemma**
estimate θ^* **vs.** get high rewards Y_t

Goal:

Maximize cumulative rewards $\longleftrightarrow$ Minimize **pseudo-regret**

$$R_T = \sum_{t=1}^T \max_{a \in \mathcal{D}_t} \langle \theta^*, a \rangle - \sum_{t=1}^T \langle \theta^*, D_t \rangle$$

Pseudo-regret: cf. $\sum_{t=1}^T Y_t$ is $\sqrt{T}$ close to $\sum_{t=1}^T \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] = \sum_{t=1}^T \langle \theta^*, D_t \rangle$

Setting: pick $D_t \in \mathcal{D}_t \subset \mathbb{R}^d$ and get $Y_t = \underbrace{\langle \theta^*, D_t \rangle}_{\text{mean effect}} + \underbrace{\eta_t}_{\text{noise}}$

Finite-armed bandits: $d = K$ and $\mathcal{D}_t = \{\mathbf{e}_1, \dots, \mathbf{e}_K\}$

$(\mathbf{e}_1, \dots, \mathbf{e}_K)$ is the canonical basis of $\mathbb{R}^K$

The mean rewards equal $\mu_a^* = \theta_a^*$

Sleeping finite-armed bandits: $d = K$ and $\mathcal{D}_t \subseteq \{\mathbf{e}_1, \dots, \mathbf{e}_K\}$

Combinatorial bandits

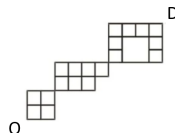
Selection of m best arms: e.g., d ads, m comparable locations

$\rightarrow \mathcal{D} = \{a \in \{0, 1\}^d : \|a\|_1 = m\}$

Shortest path: d edges, set of paths from O to D

$\rightarrow \mathcal{D} \subseteq \{0, 1\}^d$

Only the total time from O to D is observed



Setting: pick $D_t \in \mathcal{D}_t \subset \mathbb{R}^d$ and get $Y_t = \langle \theta^*, D_t \rangle + \eta_t$

Linear contextual bandits: At each round $t = 1, 2, \dots$,

0. A context $\mathbf{x}_t \in \mathbb{R}^m$ is revealed

No assumption on its generation

1. Statistician picks $A_t \in [K]$

2. She gets a reward $Y_t = \langle \theta^*, \varphi(\mathbf{x}_t, A_t) \rangle + \eta_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^* + \eta_t$
where η_t is an independent noise

3. This reward Y_t is the only feedback she receives

Comments

- $\theta^* \in \mathbb{R}^d$ is unknown
- The transfer function $\varphi : \mathbb{R}^m \times [K] \rightarrow \mathbb{R}^d$ is known
- Amounts to picking in $\mathcal{D}_t = \{\varphi(\mathbf{x}_t, a) : a \in [K]\}$
- **More concrete** formulation good for modeling $\rightarrow$ stick to it
- Pseudo-regret rewrites as:

$$R_T = \sum_{t \leq T} \max_{a \in [K]} \varphi(\mathbf{x}_t, a)^\top \theta^* - \sum_{t \leq T} \varphi(\mathbf{x}_t, A_t)^\top \theta^*$$

Setting: context $\mathbf{x}_t$, arm $A_t \in [K]$, reward $Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^* + \eta_t$

Pseudo-regret $\sum_{t \leq T} \max_{a \in [K]} \varphi(\mathbf{x}_t, a)^\top \theta^* - \sum_{t \leq T} \varphi(\mathbf{x}_t, A_t)^\top \theta^*$

Key: learn θ^* (= estimate it while playing) — intuition:

$$\sum_{s \leq t-1} \varphi(\mathbf{x}_s, A_s) Y_s \approx \sum_{s \leq t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\top \theta^* \quad \text{are } \sqrt{t}\text{-close}$$

$$\text{Thus } \theta_\star \text{ close to } \left(\sum_{s \leq t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\top \right)^{-1?} \sum_{s \leq t-1} \varphi(\mathbf{x}_s, A_s) Y_s$$

→ **LinUCB** with regularization $\lambda > 0$

Abbasi-Yadkori, Pál, Szepesvári [2011]

$$\hat{\theta}_{t-1} = \left(\lambda \text{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\top \right)^{-1} \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) Y_s$$

Setting: context $\mathbf{x}_t$, arm $A_t \in [K]$, reward $Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^* + \eta_t$

Pseudo-regret $\sum_{t \leq T} \max_{a \in [K]} \varphi(\mathbf{x}_t, a)^\top \theta^* - \sum_{t \leq T} \varphi(\mathbf{x}_t, A_t)^\top \theta^*$

LinUCB with regularization $\lambda > 0$ (Abbasi-Yadkori, Pál, Szepesvári, 2011)

Step 1: estimate

$$\hat{\theta}_{t-1} = \left(\lambda \text{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\top \right)^{-1} \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) Y_s$$

Alternative justification by ridge regression:

$$\arg \min_{\theta \in \mathbb{R}^m} \left\{ \lambda \|\theta\|^2 + \sum_{s=1}^{t-1} (Y_s - \varphi(\mathbf{x}_s, A_s)^\top \theta)^2 \right\} = \{\hat{\theta}_{t-1}\}$$

Setting: context $\mathbf{x}_t$, arm $A_t \in [K]$, reward $Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^* + \eta_t$

Pseudo-regret $\sum_{t \leq T} \max_{a \in [K]} \varphi(\mathbf{x}_t, a)^\top \theta^* - \sum_{t \leq T} \varphi(\mathbf{x}_t, A_t)^\top \theta^*$

LinUCB with regularization $\lambda > 0$ (Abbasi-Yadkori, Pál, Szepesvári, 2011)

Step 1: estimate

$$\hat{\theta}_{t-1} = \left(\lambda \text{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\top \right)^{-1} \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) Y_s$$

Step 2 [optimistic choice]: pick

$$A_t \in \arg \max_{a \in [K]} \left\{ \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{t-1} + \varepsilon_{t-1}(\mathbf{x}_t, a) \right\}$$

Confidence bonus $\varepsilon_{t-1}(\mathbf{x}_t, a)$ and regularization $\lambda > 0$ to be determined by the analysis

Reminder:
$$\hat{\theta}_{t-1} = \underbrace{\left(\lambda \text{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\top \right)^{-1}}_{=G_{t-1}} \underbrace{\sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \left(\varphi(\mathbf{x}_s, A_s)^\top \theta^* + \eta_s \right)}_{=G_{t-1} - \lambda \text{Id}_d}$$

Analysis 1/2: **determination of $\varepsilon_{t-1}(\mathbf{x}_t, a)$**

Such that $\left| \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^\top \theta^* \right| \leq \varepsilon_{t-1}(\mathbf{x}_t, a) \quad \text{w.p. } 1 - \delta$

Proof:
$$\hat{\theta}_{t-1} - \theta^* = G_{t-1}^{-1} \left(-\lambda \text{Id}_d \theta^* + \sum_{s \leq t-1} \varphi(\mathbf{x}_s, A_s) \eta_s \right) \quad \text{thus}$$

$$\begin{aligned} \left| \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^\top \theta^* \right| &\leq \left\| G_{t-1}^{-1/2} \varphi(\mathbf{x}_t, a) \right\| \left\| G_{t-1}^{1/2} (\hat{\theta}_{t-1} - \theta^*) \right\| \\ &\leq \underbrace{\left\| G_{t-1}^{-1/2} \varphi(\mathbf{x}_t, a) \right\|}_{\text{known}} \left(\underbrace{\sqrt{\lambda} \|\theta^*\|}_{\leq C} + \underbrace{\left\| G_{t-1}^{-1/2} \sum_{s \leq t-1} \varphi(\mathbf{x}_s, A_s) \eta_s \right\|}_{\text{see next slide}} \right) \end{aligned}$$

where we used $G_{t-1} \succeq \lambda \text{Id}_d$ so that $G_{t-1}^{-1/2}(\lambda \text{Id}_d) \preceq \sqrt{\lambda} \text{Id}_d$

Intermezzo: Martingales + Linear algebra

Theorem (Abbasi-Yadkori, Pál, Szepesvári, 2011)

Filtration $(\mathcal{F}_s)_{s \geq 0}$ *Scalar-valued $(\eta_s)_{s \geq 1}$ adapted & σ^2 -sub-Gaussian cond. to $\mathcal{F}_{s-1}$* *d -vector-valued $(X_s)_{s \geq 1}$ predictable & $\|X_s\| \leq 1$*

Let $M_t = \sum_{s \leq t} \eta_s X_s$ and $G_t = \lambda \text{Id}_d + \sum_{s \leq t} X_s X_s^\top$

W.p. $1 - \delta$, for all $t \geq 1$,

$$\|G_t^{-1/2} M_t\| \leq \sigma \sqrt{2 \ln(1/\delta) + d \ln(1 + t/(\lambda d))} = \mathcal{O}(\sqrt{\ln(t/\lambda)})$$

Application: $X_s = \varphi(\mathbf{x}_s, A_s)$ & cond. to $\mathcal{F}_{s-1} \leftrightarrow$ integr. randomness η_s

Intermezzo: Martingales + Linear algebra

1. Laplace's method for $f : [a, b] \rightarrow \mathbb{R}$ with single max. at u_0

$$I_x = \int_a^b \exp(x f(u)) du = \exp(x f(u_0)) \underbrace{\int_a^b \exp(x (f(u) - f(u_0))) du}_{\leq \text{constant}}$$

$$\text{thus} \quad f(u_0) \approx \frac{\ln I_x}{x} \quad \text{for large } x > 0$$

2. To be used over the supermartingales (cf. sub-Gaussian noises)

$$E_t(x) = \exp\left(\sum_{s \leq t} x^\top X_s \eta_s - \frac{\sigma^2}{2} (x^\top X_s)^2\right) = \exp\left(x^\top M_t - (\sigma^2/2) x^\top G_t x\right)$$

Intermezzo: Martingales + Linear algebra

Supermartingales $E_t(x) = \exp\left(x^\top M_t - (\sigma^2/2) x^\top G_t x\right)$

Supermartingale $\bar{E}_t = \int_{\mathbb{R}^d} E_t(x) dx$

$$= [\dots] = \sqrt{\det(2\pi\sigma^{-2} G_t^{-1})} \exp\left(\frac{1}{2\sigma^2} \|G_t^{-1/2} M_t\|^2\right)$$

Cf. formulas for Gaussian densities

Doob: $\mathbb{E}[\bar{E}_t] \leq \mathbb{E}[\bar{E}_0] = [\dots] = \sqrt{\det(2\pi\lambda^{-1} \text{Id}_d)}$

then Markov:

$$\mathbb{P}\left(\|G_t^{-1/2} M_t\| > u\right) \leq \exp\left(-\frac{u^2}{2\sigma^2}\right) \mathbb{E}\left[\frac{1}{2\sigma^2} \exp\left(\|G_t^{-1/2} M_t\|^2\right)\right]$$

Also bound $\det(G_t) \leq (\lambda + t/d)^d$

$$Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^* + \eta_t \quad \text{and} \quad \hat{\theta}_{t-1} = \underbrace{\left(\lambda \text{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\top \right)^{-1}}_{=G_{t-1}} \underbrace{\sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \left(\varphi(\mathbf{x}_s, A_s)^\top \theta^* + \eta_s \right)}_{=G_{t-1} - \lambda \text{Id}_d}$$

Analysis 1/2: **determination of $\varepsilon_{t-1}(\mathbf{x}_t, a)$**

Such that $\left| \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^\top \theta^* \right| \leq \varepsilon_{t-1}(\mathbf{x}_t, a) \quad \text{w.p. } 1 - \delta$

Proof: so far, w.p. $1 - \delta$,

$$\begin{aligned} \left| \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^\top \theta^* \right| &\leq \left\| G_{t-1}^{-1/2} \varphi(\mathbf{x}_t, a) \right\| \left\| G_{t-1}^{1/2} (\hat{\theta}_{t-1} - \theta^*) \right\| \\ &\leq \underbrace{\left\| G_{t-1}^{-1/2} \varphi(\mathbf{x}_t, a) \right\|}_{\text{known}} \left(\underbrace{\sqrt{\lambda} \|\theta^*\|}_{\leq C} + \underbrace{\left\| G_{t-1}^{-1/2} \sum_{s \leq t-1} \varphi(\mathbf{x}_s, A_s) \eta_s \right\|}_{\text{known bound } \mathcal{O}(\sqrt{\ln(t/\lambda)})} \right) \end{aligned}$$

→ we read our $\varepsilon_{t-1}(\mathbf{x}_t, a)$

$(1 - \delta)$ -confidence region on θ^* : $\left\{ \theta : \left\| G_{t-1}^{1/2} (\hat{\theta}_{t-1} - \theta) \right\| \leq \sqrt{\lambda} C + \mathcal{O}(\sqrt{\ln(t/\lambda)}) \right\}$

$$\hat{\theta}_{t-1} \quad \text{s.t.} \forall a \in [K], \quad \left| \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{t-1} - \varphi(\mathbf{x}_t, a)^\top \theta^* \right| \leq \varepsilon_{t-1}(\mathbf{x}_t, a) = \left\| G_{t-1}^{-1/2} \varphi(\mathbf{x}_t, a) \right\| \left(C\sqrt{\lambda} + \mathcal{O}(\sqrt{\ln(t/\lambda)}) \right)$$

$$\longrightarrow \text{Pick} \quad A_t \in \arg \max_{a \in [K]} \left\{ \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{t-1} + \varepsilon_{t-1}(\mathbf{x}_t, a) \right\} \quad \text{and get} \quad Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^* + \eta_t$$

$$\text{Aim: control pseudo-regret} \quad \sum_{t \leq T} \max_{a \in [K]} \varphi(\mathbf{x}_t, a)^\top \theta^* - \sum_{t \leq T} \varphi(\mathbf{x}_t, A_t)^\top \theta^*$$

Analysis 2/2: $\sqrt{T}$ -gap-free regret bound

Proof: $-\varphi(\mathbf{x}_t, A_t)^\top \theta^* \leq \varepsilon_{t-1}(\mathbf{x}_t, A_t) - \varphi(\mathbf{x}_t, A_t)^\top \hat{\theta}_{t-1} \quad \text{and}$

$$\begin{aligned} \max_{a \in [K]} \varphi(\mathbf{x}_t, a)^\top \theta^* &\leq \max_{a \in [K]} \left\{ \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{t-1} + \varepsilon_{t-1}(\mathbf{x}_t, a) \right\} \\ &= \varphi(\mathbf{x}_t, A_t)^\top \hat{\theta}_{t-1} + \varepsilon_{t-1}(\mathbf{x}_t, A_t) \quad \text{cf. def. } A_t \end{aligned}$$

$$\begin{aligned} \text{Thus,} \quad \sum_{t \leq T} \max_{a \in [K]} \varphi(\mathbf{x}_t, a)^\top \theta^* - \sum_{t \leq T} \varphi(\mathbf{x}_t, A_t)^\top \theta^* &\leq 2 \sum_{t \leq T} \varepsilon_{t-1}(\mathbf{x}_t, A_t) \\ &\leq \underbrace{\left(C\sqrt{\lambda} + \mathcal{O}(\sqrt{\ln(T/\lambda)}) \right)}_{\leq \tilde{\mathcal{O}}(1)} \underbrace{\sum_{t \leq T} \left\| G_{t-1}^{-1/2} \varphi(\mathbf{x}_t, A_t) \right\|}_{\leq \tilde{\mathcal{O}}(\sqrt{T})} \quad \text{see next slide} \end{aligned}$$

Remains to show

Dealing with $\varphi(\mathbf{x}_t, \mathbf{A}_t)$ is vital!

$$\sum_{t \leq T} \left\| G_{t-1}^{-1/2} \varphi(\mathbf{x}_t, \mathbf{A}_t) \right\| \leq \tilde{O}(\sqrt{T})$$

Elliptic potential lemma (Abbasi-Yadkori, Pál, Szepesvári, 2011)

$$G_{t-1} = \overbrace{\lambda \text{Id}_d + \sum_{s \leq t-1} X_s X_s^\top}^{\succeq \lambda \text{Id}_d \succeq \text{Id}_d} \quad \text{where} \quad X_s = \varphi(\mathbf{x}_s, \mathbf{A}_s)$$

$$\begin{aligned} \sum_{t \leq T} \left\| G_{t-1}^{-1/2} X_t \right\| &\leq \sqrt{T \sum_{t \leq T} X_t^\top G_{t-1} X_t} && \text{C-Schw. \& } X_t^\top G_{t-1} X_t \leq \|X_t\|^2 \leq 1 \\ &\leq \sqrt{2T \sum_{t \leq T} \ln(1 + X_t^\top G_{t-1} X_t)} && u \leq \ln(1 + u) \text{ for } u \leq 1 \\ &= \sqrt{2T \frac{\ln(\det(G_T))}{\ln(\det(\lambda \text{Id}_d))}} && \text{equality: "matrix-determinant lemma"} \\ &\leq \sqrt{2T} \tilde{O}(1) \end{aligned}$$

UCB as a special case of LinUCB

Vanilla K -armed bandits $\leftrightarrow \varphi : (\mathbf{x}, a) \mapsto \mathbf{e}_a$ and $\theta^* = (\mu_a)_{a \in [K]}$

$$Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^* + \eta_t = \mu_{A_t} + \eta_t$$

$$\begin{aligned} \hat{\theta}_{t-1} &= \overbrace{\left(\lambda \text{Id}_d + \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) \varphi(\mathbf{x}_s, A_s)^\top \right)}^{= G_{t-1} = \text{diag}((\lambda + N_a(t-1))_{a \in [K]})}^{-1} \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, A_s) Y_s \\ &= \left(\frac{1}{\lambda + N_a(t-1)} \sum_{s \leq t-1} Y_s \mathbb{I}_{\{A_s=a\}} \right)_{a \in [K]} = (\hat{\mu}_a(t-1))_{a \in [K]} \quad \text{when } \lambda = 0 \end{aligned}$$

Similar confidence bonuses proportional to

$$\left\| G_{t-1}^{-1/2} \varphi(\mathbf{x}_t, a) \right\| \sqrt{\ln t} = \sqrt{\frac{\ln t}{\lambda + N_a(t-1)}}$$

Thus, similar $\tilde{\mathcal{O}}(\sqrt{T})$ gap-free regret bounds

Linear contextual bandits: optimality

See Lattimore and Szepesvári [2020, Chapter 25]

The $\tilde{O}(\sqrt{T})$ gap-free bound exhibited for LinUCB is fine

In T gap-dependent bounds may be achieved with a different class of algorithms

- not based on optimism
- performing tracking à la Graves and Lai [1997]

Logistic bandits

Extended from Faury, Abeille, Calauzènes, Fercoq [2020] See GLMs in Filippi, Cappé,
Garivier, Szepesvári [2010]

At each round $t = 1, 2, \dots$,

0. A context $\mathbf{x}_t \in \mathbb{R}^d$ is determined by the environment
1. Statistician picks arm $A_t \in [K]$
2. The **outcome** $Y_t \in \{0, 1\}$ is drawn with probability $P(\mathbf{x}_t, A_t)$
This is the only feedback Statistician receives
3. Statistician gets the reward $r(\mathbf{x}_t, A_t) Y_t$

Conversion rate P unknown but reward function r known

Structural assumption:

$$P(\mathbf{x}, a) = \eta(\varphi(\mathbf{x}, a)^\top \theta_\star) \quad \text{where} \quad \eta(x) = \frac{1}{1 + e^{-x}}$$

Similar results may be achieved as for linear bandits

Estimation based on maximum likelihood

Linear contextual bandits with latent spates

See Nelson et al. [2022], Li and Stoltz [2026]

Latent states $h_t \in [H]$ e.g., economic state

Rewards $Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta_{h_t}^* + \eta_t \rightarrow (\theta_h^*)_{h \in [H]} \in \mathbb{R}^{dH}$

Hidden Markov model: $\hat{\mathbf{b}}_t(h)$ estimates $\mathbf{b}_t(h) = \mathbb{P}(h_t = h \mid \mathbf{x}_{1:t})$

LinUCB strategy on estimated beliefs:

- Estimate $(\hat{\theta}_{t-1,h})_{h \in [H]} = G_{t-1}^{-1} \sum_{s=1}^{t-1} (\hat{\mathbf{b}}_s(h) \varphi(\mathbf{x}_s, A_s))_{h \in [H]} Y_s$

- Pick

$$A_t \in \arg \max_{a \in [K]} \left\{ \sum_{h \in [H]} \hat{\mathbf{b}}_t(h) \varphi(\mathbf{x}_t, a)^\top \hat{\theta}_{\dots, h} + \varepsilon_{t-1}(\mathbf{x}_t, a) \right\}$$

Complex statistical dependencies: A_t and $(h_s)_{s \leq t}$ are linked through the $(Y_s)_{s \leq t}$
→ HMM forgetting conditions + Proceed in stages: $\hat{\theta}_{s_t-1, h}$

Toy application

Balance between production and demand: toy setting

As electricity is hard to store, balance between production and demand must be strictly maintained



Adapt production

and

Forecast demand

Optimization

Statistics

To achieve a balance between production and demand



We exploit the following sources of flexibility:

